Nightreign Recognizer Evaluation

Assessment of the proto-authoritative pair, math choices, precision/performance alignment, and current GUI issue

Component	Alignment score	Notes
recognize_nightreign_map_proto_v12_authoritative.py	84%	Strongly aligned with the recent precision fixes.
nightreign_map_recognizer_gui_hybrid_v25_proto_authoritative.py	58%	Correct direction, but structurally fragile and missing helper glue in the delivered file.
Pair together	71%	Logic direction is good, but the GUI deliverable still needs cleanup to be dependable.

1. Executive summary

The authoritative proto file mostly follows the direction that recent debugging demanded: normalize the panel before recognition, sanitize unstable scores, refine weak fallback crops using Shifting Earth confidence, and only then classify slots and score layouts. That is why its score is high.

The GUI file aims in the right direction but is less trustworthy as a deliverable because it still mixes legacy GUI logic with a second, proto-backed path. The attached runtime error confirms that the delivered GUI file was missing helper glue needed by that path. That lowers the confidence score for the GUI specifically.

The latest monitor tests suggest that the crop-refinement step was the biggest recent win. After refinement, monitor 2 converged to the same earth family and almost the same ranked layouts as monitor 1. That is a strong sign that the math is solving the right problem.

2. What still makes sense

Canonical panel normalization. This is the right answer to the 4K vs 1080 and 150% vs 100% scaling problem. It reduces the dependence of slot matching on raw monitor resolution.

Crop refinement driven by earth score. Using a small search over crop nudges and picking the crop with the best Shifting Earth objective is mathematically sensible and empirically effective.

Finite-score sanitization. Earlier runs produced Infinity and NaN values. The later runs no longer showed that failure mode, so this change should stay.

Context refinement only for unknown slots. Promoting unknowns only when the top layouts agree is a conservative and reasonable second-stage heuristic.

3. What is still weak

There are still many **unknown** slot labels even in the improved runs. That means the current main bottleneck is not crop alignment anymore; it is the local landmark classifier inside each slot window.

The GUI file is still structurally patchy. It should eventually be reduced to one clean recognition path instead of carrying both older and newer recognition logic in the same file.

The context-refinement step is useful, but it is still heuristic. If the top layouts are wrong, it can reinforce the wrong answer. That means it should remain conservative.

4. Math evaluation

Shifting Earth score. The current weighted combination of low-frequency terrain similarity and gradient similarity is a good choice for the problem. Terrain identity is mostly about broad landmass structure, not about individual icons.

```
earth_score = 0.65 * low_corr + 0.35 * grad_corr
```

Crop refinement objective. The objective rewards a confident best earth, rewards a healthy margin over the second-best earth, penalizes large black borders, and mildly rewards edge-rich crops. This is practical rather than elegant, but appropriate for the task.

```
objective =
    top_earth_score
  + 0.35 * (top_earth_score - second_earth_score)
  - 0.12 * black_ratio
  + 0.03 * edge_mean
```

Layout score. The current scoring is heuristic but coherent: reward the expected label score, subtract strong alternative-label evidence, and penalize empty slots that look occupied. It is not a formal probabilistic model, but it is internally consistent.

Context refinement. The exponential reweighting of top layouts is reasonable because it lets close runners-up influence unknown-slot promotion without giving equal weight to poor matches.

```
weight_i = exp((score_i - top_score) / 3.0)
```

Diagram A - recognition flow

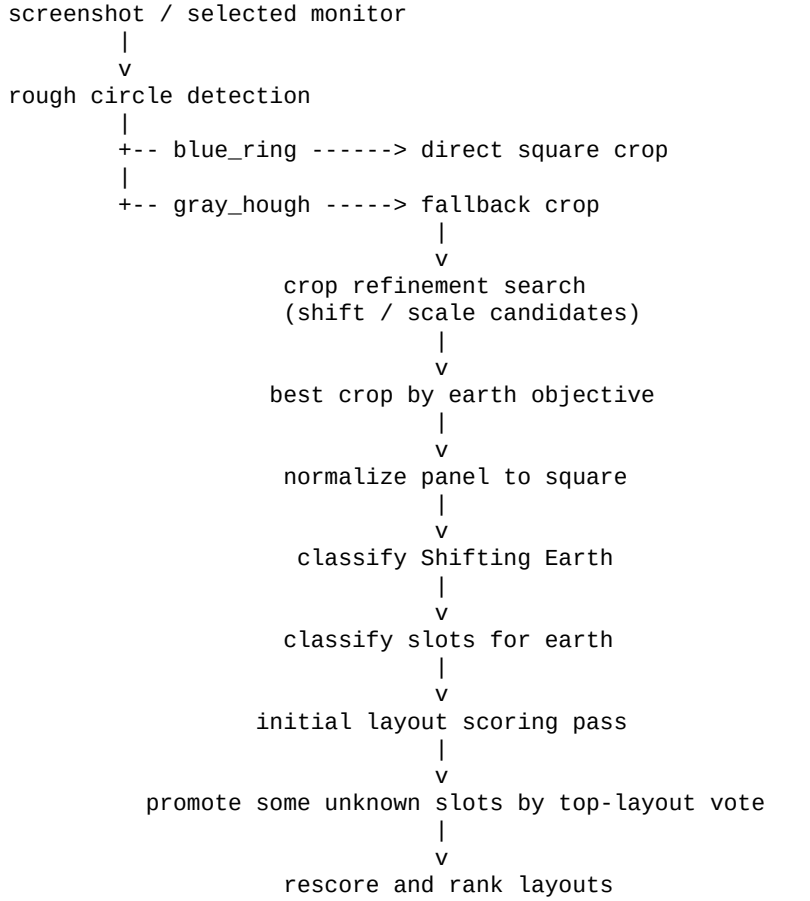
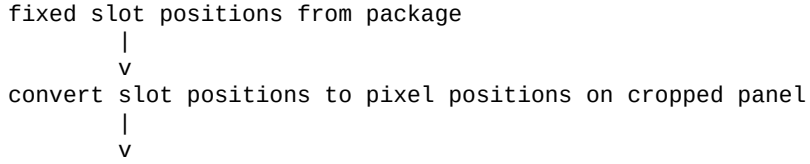


Diagram B - why a slightly wrong crop hurts



```
sample local patch around each slot

If crop is correct:
    slot window overlaps true landmark

If crop is shifted:
    slot window samples nearby terrain / border / wrong icon
    -> many labels become unknown or wrong
```

Diagram C - crop refinement search

```
initial crop box
   |
   +-- dx in { -4%, -2%, 0, +2%, +4% }
   +-- dy in { -4%, -2%, 0, +2%, +4% }
   +-- scale in { 0.97, 1.00, 1.03 }

for each candidate:
    normalize panel
    compute earth ranking
    compute black_ratio and edge_mean
    compute objective

pick candidate with max objective
```

5. Interpretation of the current GUI error

The error **name "load_proto_backend_module" is not defined** is not a mathematical problem. It is a file-assembly problem in the delivered GUI script. The file tries to call helper functions that were not actually included in the final script body.

That means the evaluation score for the GUI should be read with this caveat: the intended recognition direction is decent, but the delivered file needs patching before it can be trusted operationally.

A fixed GUI file should include two small helpers: one that loads the authoritative proto backend module and one that resolves the slot package path. Those helpers are glue code, but without them the GUI path cannot run.

6. Recommendation

Keep the **proto math** as the current source of truth. Keep crop refinement, normalization, finite-score sanitization, and conservative context refinement. The next best improvement target is not more crop math; it is reducing the number of unknown slot labels without over-promoting weak matches.

Separately, clean the GUI file into a single path that delegates to the proto backend reliably. Right now the recognition math is in better shape than the GUI plumbing.